README

Projection-enumerators

This project contains all R codes and results for the functions, simulations, and examples in the paper "A General Theory of Projection Enumerators for Factorial and Space-Filling Designs". Each file is described below.

1 PSE.R

This file provides four functions for a given design D with levels s^p.

- PSE(x,y,D,s,p): computes $E_\chi(D; x, y)$.
- TPSE(x,y, D,s,p,t): computes the truncated enumerator $E_\chi^{(t)}(D; x, y)$.
- PSFPI(D,s,p): quickly computes all all $P_{i,j}(D)$.
- TPSFPI(D,s,p,t): quickly computes $P_{i,j}(D)$ for $j \leq t$.

2 PBE.R

Similar to "PSE.R", but designed for $B_{i,j}(D)$ when design D has s levels.

- PBE(x,y,D,s): computes $E_\varphi(D; x, y)$.

- TPBE(x,y,D,s,t): computes the truncated enumerator $E_{\varphi}^{(t)}(D; x, y)$.
- PWPI(D,s): quickly computes all all $B_{i,j}(D)$.
- TPWPI(D,s,t): quickly computes $B_{i,j}(D)$ for $j \leq t$.

3 GSOA_opt.R

Contains the function, "GSOA_opt(polyprimitive, s, K = (s - 1), q = (length(polyprimitive)-1), add = 0)" to construct the optimal GSOA($s^p, K * (s^p - 1)/(s - 1), s^q, t$). We need to illustrate the inputs "polyprimitive", "q", and "add".

- polyprimitive: It is a coefficient vector to determine the primitive polynomial. For instance, if we are going to construct a design with 3^2 levels with the primitive polynomial $x^2 + x + 2$, then the vector should be $c(1, 1, 2)$. If we are going to construct a design with 3^3 levels with the primitive polynomial $x^3 + 2x + 1$, then the vector should be $c(1, 0, 2, 1)$.
- q: It determines the number of levels as s^q. By default, the level count is s^p (from the polynomial). For example, if the polynomial corresponds to $3^3 = 27$ but you set, then you obtain a GSOA with $3^2 = 9$ levels.
- add: It is a shift vector (default 0, meaning no shift). Shifting does not affect distance properties, so shifted GSOAs remain optimal. We explain the use of "add" by two examples. If we want to shift the GSOA by 13 when $s = 3$, we should express 13 using 3-base, that is, $13 = 1 \times 3^2 + 1 \times 3 + 1$, thus "add" is "c(1,1,1)". Similarly, if we want to shift the GSOA by 11 when $s = 5$, we should express 11 using 5-base, that is, $11 = 2 \times 5 + 1$, thus "add" is set as "c(2,1)".

4 stratification_pattern.R

This file comes from Shi and Xu (2024)'s codes, and the function "SP" computes the projection stratification pattern "$P_{i,j}(D)$" with 2^3 levels using the definition of the projection stratification pattern.

5 Examples, Simulations and Comparisons

These folders contain all code and results for examples, simulations, and comparisons. Notice that some scripts require functions from "PSE.R", "PBE.R", or "GSOA_opt.R", so you may should put the necessary files in the same path. In "comparisons", "selext_good_design.R" selects the best shift of a GSOA under a distance criterion. In "simulations", the file "functions_test" provides all test functions used in the simulations.